457.568 Statistical Hydrology, Fall 2022

R Code Training Assignment #2 (Extreme Event Analysis)

Due before the lecture of Nov-15, 2022

<Notice> Include responses to every question below, and make sure your final assignment is well organized (i.e., numbered responses, full sentences, code attached at the end). This makes it easier to grade & to give partial credits as well.

1. Conduct the extreme event analysis using the data provided as "fallcreek.csv" following the steps below.

   A. Load the data into R using the read.csv() function into a variable in R called FallCreek. Open the "fallcreek.csv" file to see how the data is organized. Save the flow of the "fallcreek.csv" file into a variable named 'AnnMaxFlow.'

   B. It is always helpful to visualize the annual maxima as a time series against water year. However, to do this, we first need to manipulate the original data, specifically the dates. Using the mutate verb in the dplyr package, change the Date column to be a Date object in R (hint: you can use the as.Date() function within the mutate verb).

   C. Using the ggplot2 package, create a line graph between the date and the annual maxima, as well as two histograms of the annual maxima, one on the original data scale and another on the log10 scale. Comment on the shape of the distribution on both scales.

   D. We are going to examine the fit of a GEV distribution to the annual maxima using the "ismev" library in R. After installing and loading the library, you can fit a GEV distribution to your data as follows:

      my.gev.fit <- gev.fit(FallCreek$AnnMaxFlow)

      The MLE parameter estimates can be recovered with my.gev.fit$mle. Report your estimated parameters (location, scale, shape).

   E. Using ggplot, examine how well the GEV fits the data using a probability plot. To do this, use dplyr and mutate to add two new columns to FallCreek: plotting.pos and gev.cdf.

      i. plotting.pos will be the plotting positions, and can be calculated using the rank() function on the AnnMaxFlow data

      ii. gev.cdf will be the cdf of the fitted GEV evaluated on your data, and can be calculated using the GEV cdf function gevf() in the ismev package. There are two arguments to the function: a vector of 3 parameters values, and a vector of data to pass through the cdf.

      Create a probability plot of plotting positions against model-based cdf values. Include a 1:1 line (Google how for ggplot). Comment on how well the GEV distribution fits your data?

   F. Develop point estimates of the following return period events: 2-year, 10-year, 25-year, 50-year, 100-year, 200-year. To do this:

i. Create a new data.frame FallCreekDesignEvents with a single column named return.period" equal to the vector of return levels listed above.

ii. Using dplyr and mutate, create a new column ("exceed.prob") of exceedance probabilities.

iii. Use mutate to create another column ("return.level") of the design event values. Here, you can use your fitted parameter estimates and the GEV quantile function, which is conveniently provided by the gevq() function. The gevq() function takes two arguments: a vector of 3 parameter values, and a vector of exceedance probabilities.

G. Report a table of return periods and return levels. Using ggplot, develop a return period plot using these design event estimates. Make sure that the x-axis is on a log10 scale and connect your estimates with a line.

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2. In this exercise we are going to revisit the calculation of Fall Creek design floods based on annual maxima, but this time we'll compare results from the GEV and Log-Pearson Type 3 (LP3) distributions. In the later part of the exercise, you will use the Generalized Pareto distribution (GPD) for your analysis.

A. First, add onto the return period plot from 1-G in a different color estimates of design events using a log Pearson type III distribution.

i. You can find a frequency factor table as an attachment. Feel free to round your skewness estimate to the nearest tenths place.

ii. Use the mutate verb to add the LP3-based return levels to the FallCreekDesignEvents data frame before plotting.

B. Add the empirical return periods against the annual maxima on the same plot.

i. Empirical return periods can be derived from empirical non-exceedance probabilities, which were stored in the FallCreek data frame in Problem 1 to create a P-P plot. (In other words, Emp.return.period = 1/(1-plotting.pos))

C. In ggplot(), a slightly different syntax is needed if you want to plot data from multiple data frames (e.g., FallCreekDesignEvents and FallCreek) on the same plot. In the example below, two different lines are added from data in one data frame (called df1), and points are added from a different data frame (called df2). The color argument in the aesthetics is used to add labels to the legend for each plotting feature.

```
ggplot() +
geom_line(data=df1, aes(x =, y =, color = "myline1")) +
geom_line(data=df1, aes(x = ,y = ,color = "myline2")) +
geom_point(data=df2, aes(x =, y = , color = "mypoints"))
```

Make sure all of the elements on your plot are clearly labeled in the legend. You can use the same label for the upper and lower bounds of the GEV return levels. Comment on how similar or different the return level estimates are between the GEV and LP3 models, and how the observed data compares to the models.